

Lecture 7

Linear Regression & Regularization

Assigned reading:

- See last lecture's readings

Announcement

- Some typos on slides from geometric view were fixed and re-posted within L8—please redownload.

Bug fix from last class

[Linear Algebra] Matrix-Vector Multiplication

- Approach 1: So far, we've thought of our model as horizontally stacked predictions per datapoint:

$$\begin{bmatrix} | \\ \mathbb{Y} \\ | \end{bmatrix} = \begin{bmatrix} \text{---} & \mathbf{x}_1^T & \text{---} \\ \text{---} & \mathbf{x}_2^T & \text{---} \\ & \vdots & \\ \text{---} & \mathbf{x}_N^T & \text{---} \end{bmatrix} \begin{bmatrix} | \\ \mathbf{w} \\ | \end{bmatrix}$$

Matrix sizes:

$$N \times 1 = (N \times d) (d \times 1)$$

$$\mathbf{Y} = \mathbf{A}\mathbf{w} = \mathbb{X}\mathbf{w}$$

- Approach 2: Helpful to think of matrix-vector multiplication as a **linear combination of feature vectors**, scaled by **parameters**.

$$\begin{bmatrix} | \\ \mathbb{Y} \\ | \end{bmatrix} = \begin{bmatrix} | & & | \\ \mathbb{X}_{:,1} & \dots & \mathbb{X}_{:,d} \\ | & & | \end{bmatrix} \begin{bmatrix} | \\ \mathbf{w} \\ | \end{bmatrix} = w_1 \mathbb{X}_{:,1} + w_2 \mathbb{X}_{:,2} + \dots + w_d \mathbb{X}_{:,d} = \sum_j w_j \mathbb{X}_{:,j}$$

Bug fix from last class

Finding Optimum Model parameters ($\mathbf{w}$)

Vector $\vec{e}$ is orthogonal to the $\text{span}(\mathbb{X})$, meaning that:

$$\mathbb{X}^T \vec{e} = 0$$

$$\mathbb{X}^T (\mathbf{y} - \mathbb{Y}) = 0$$

$$\mathbb{X}^T \mathbf{y} - \mathbb{X}^T \mathbb{X} \mathbf{w}^* = 0$$

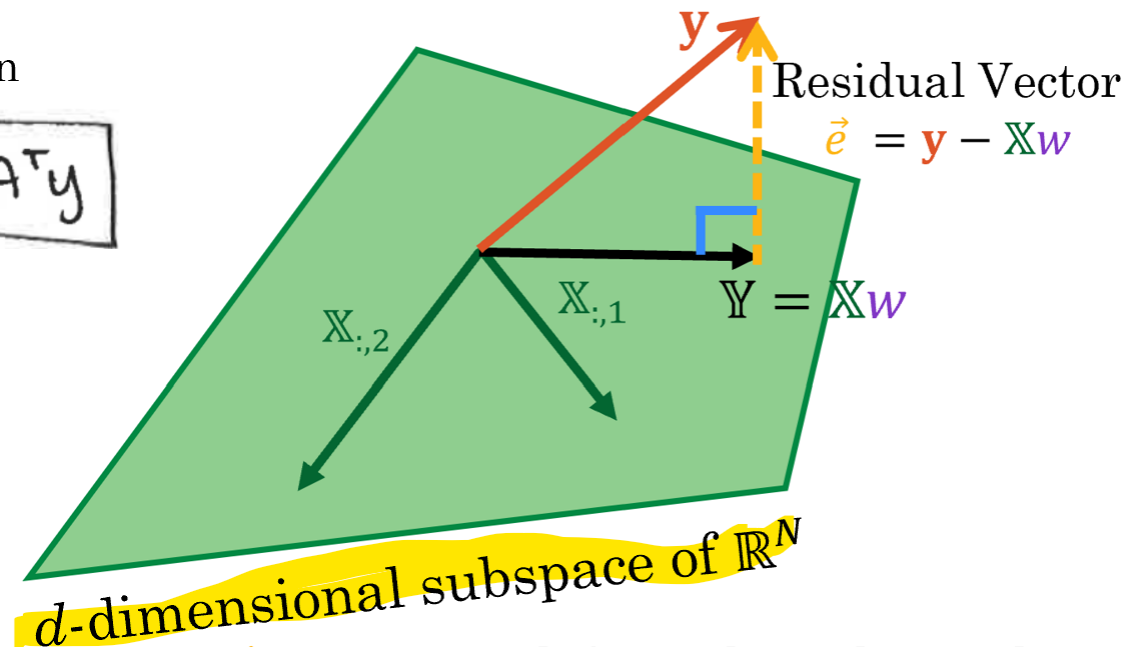
$$\mathbb{X}^T \mathbf{y} = \mathbb{X}^T \mathbb{X} \mathbf{w}^*$$

$$\Rightarrow \mathbf{w}^* = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbf{y}$$

If $\mathbb{X}^T \mathbb{X}$ is invertible

MLE solution

$$\mathbf{w} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$

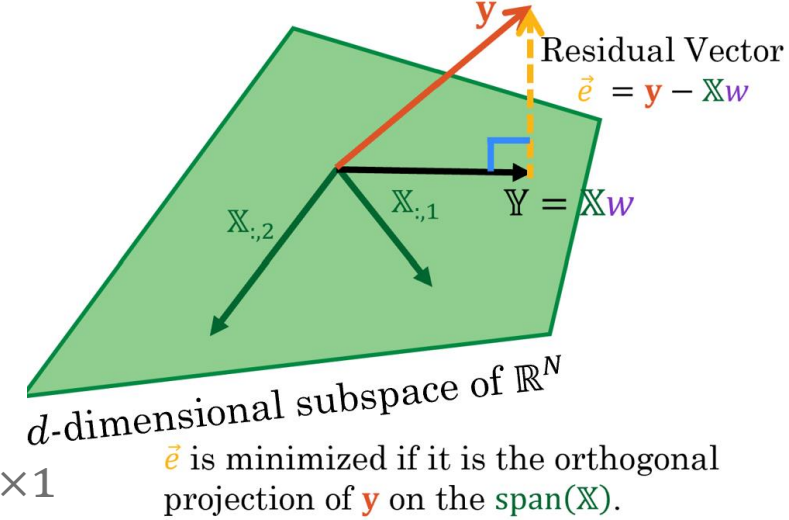


$\vec{e}$ is minimized if it is the orthogonal projection of $\mathbf{y}$ on the $\text{span}(\mathbb{X})$.

Recap from last class:

From both the MLE and geometric view:

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad \mathbf{X} \in \mathbb{R}^{n \times d}, \mathbf{y} \in \mathbb{R}^{n \times 1}, \mathbf{w}^* \in \mathbb{R}^{d \times 1}$$



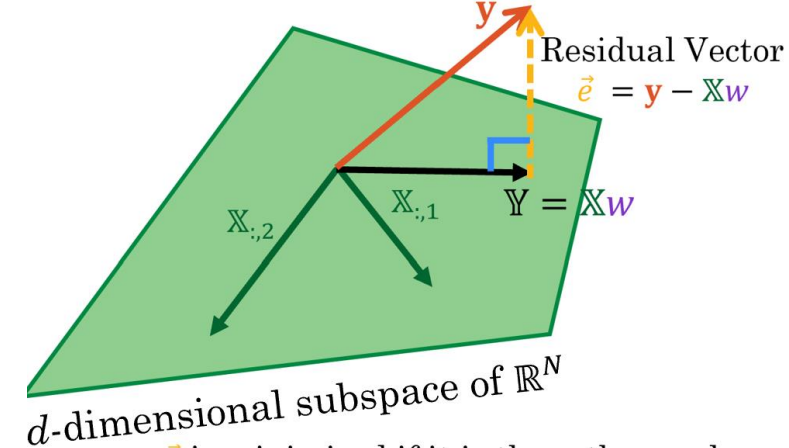
when $\mathbf{X}^T \mathbf{X} \in \mathbb{R}^{d \times d}$ is invertible, i.e., $\text{rank}(\mathbf{X}^T \mathbf{X}) = d$.

(i.e., not invertible when any features are dependent, which is guaranteed to happen when there are more features than samples, but could happen with fewer features.)

- If $\mathbf{X}^T \mathbf{X}$ is not invertible, can still use the pseudo-left inverse (Moore-Penrose inverse, $\mathbf{X}^+$), $\mathbf{w}^* = \mathbf{X}^+ \mathbf{y}$.
- Of the infinite solutions $\mathbf{w}^*$ that yield minimal error, $\|\mathbf{X}\mathbf{w}^* - \mathbf{y}\|^2$, Moore-Penrose will give the one with the smallest $\|\mathbf{w}^*\|_2$.

Recap from last class:

From both the MLE and geometric view:



$$w^* = (X^T X)^{-1} X^T y \quad X \in \mathbb{R}^{n \times d}, y \in \mathbb{R}^{n \times 1}, w^* \in \mathbb{R}^{d \times 1}$$

$\vec{e}$ is minimized if it is the orthogonal projection of y on the $\text{span}(X)$.

when $X^T X \in \mathbb{R}^{d \times d}$ is invertible, i.e., $\text{rank}(X^T X) = d$.
(i.e., not invertible when any features are dependent, which is good when there are more features than samples, but could happen with

$$\begin{aligned} -A^T y + A^T A w &= 0 \\ A^T A w &= A^T y \\ w &= (A^T A)^{-1} A^T y \end{aligned}$$

- If $X^T X$ is not invertible, can still use the pseudo-left inverse (Penrose inverse, X^+), $w^* = X^+ y$.
- Of the infinite solutions w^* that yield minimal error, $\|Xw^* - y\|^2$, Moore-Penrose will give the one with the smallest $\|w^*\|_2$.

What is this Moore-Penrose Inverse?

First, recall the *Singular Value Decomposition (SVD)*:

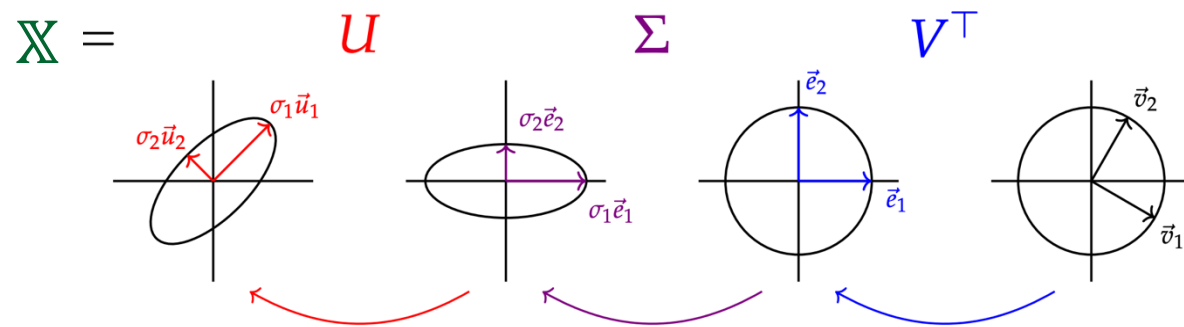
- Every real matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ can be factored as

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \mathbf{U} \in \mathbb{R}^{n \times n}, \mathbf{\Sigma} \in \mathbb{R}^{n \times d}, \mathbf{V} \in \mathbb{R}^{d \times d}$$

- where $\mathbf{U}$ and $\mathbf{V}$ are rotations and have orthonormal columns ($\mathbf{U}^T\mathbf{U} = \mathbf{V}^T\mathbf{V} = \mathbf{I}$).

- $\mathbf{\Sigma}$ is diagonal scaling by singular values σ_i .

- $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r, 0, 0)$ with $r = \text{rank}(\mathbf{X})$.



<https://eecs16b.org/notes/sp24/note15.pdf>

- If $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ is invertible, then $\mathbf{X}^{-1} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T$.

- If some $\sigma_i = 0$, we cannot invert directly.

- Moore-Penrose Pseudo-inverse, “flip” only the non-tiny singular values:

$$\sigma_{ii}^+ = \begin{cases} \frac{1}{\sigma_i} & \text{if } \sigma_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

Pseudo-Inverse: $\mathbf{X}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^T$

- Linear Regression: $\mathbf{w}^* = \mathbf{X}^+\mathbf{y}$.
- ALWAYS works, even with linearly dependent features.

More on the Moore–Penrose Pseudo-Inverse

Four Moore–Penrose conditions:

1. $XX^+X = X$

2. $X^+XX^+ = X^+$

3. $(XX^+)^T = XX^+$

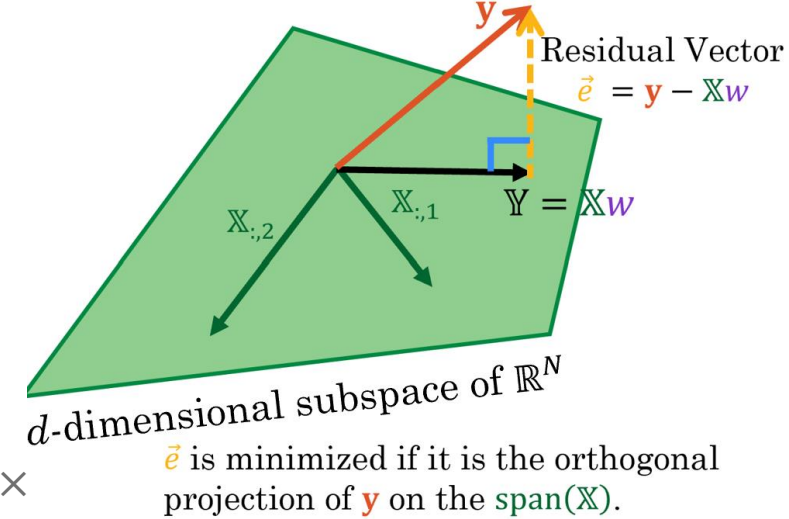
4. $(X^+X)^T = X^+X$

These conditions uniquely define the Moore–Penrose pseudoinverse, X^+ for X .

Recap from last class:

From both the MLE and geometric view:

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad \mathbf{X} \in \mathbb{R}^{n \times d}, \mathbf{y} \in \mathbb{R}^{n \times 1}, \mathbf{w}^* \in \mathbb{R}^{d \times 1}$$



when $\mathbf{X}^T \mathbf{X} \in \mathbb{R}^{d \times d}$ is invertible, i.e., $\text{rank}(\mathbf{X}^T \mathbf{X}) = d$.

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Recap from last class:

Intuition for choosing one specific $\hat{\mathbf{w}}$.

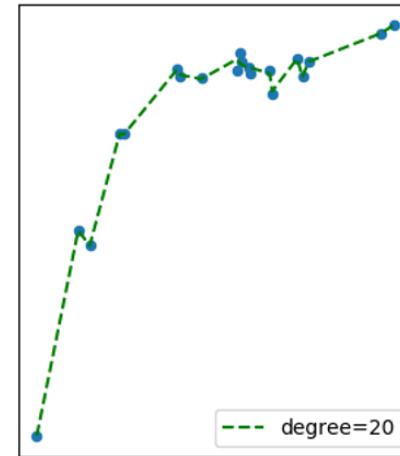
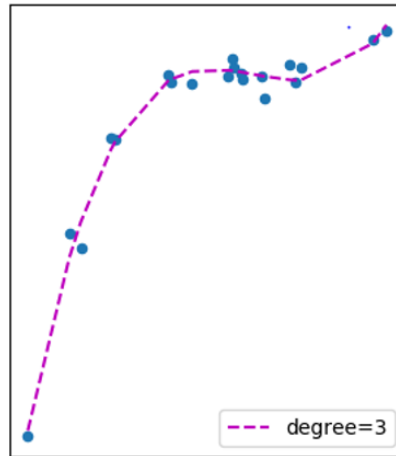
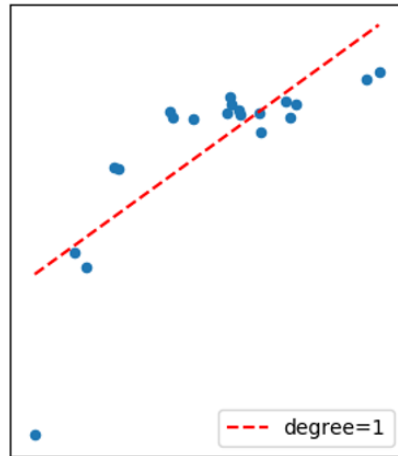
Of the ∞ solutions for $\mathbf{w}_{MLE}$, choose the one with the least norm, $\|\mathbf{w}_{MLE}\|_2$.
Why might this be a good idea? $\hat{\mathbf{y}} = \mathbf{x}^T \hat{\mathbf{w}}$

- We would like each feature to have as little effect on the outcome as possible while still predicting well, so that model behaves “gracefully”.
- Consider prediction, $\hat{\mathbf{y}} = \mathbf{w}^T \mathbf{x}$. How much does the prediction change when we perturb, $\mathbf{x}' = \mathbf{x} + \delta$, for different norm $\mathbf{w}$?
- With smaller coefficients the model is less sensitive to noisy data.
- What about for non-degenerate linear regression ($\mathbf{A}^T \mathbf{A}$ is invertible)?
- Yes! For many problems (and models), small param norm is a good idea.
- Small norm is one *e.g.*, of *regularization*:
 - in effect, reduce # free parameters, while keeping the same set of parameters!

Recap from last class:

What can go wrong in linear regression

- Even when we don't perfectly fit the training data, we are still in danger of **fitting to noise** instead of signal (causing worse predictions on test points).
- Our goal is not to fit a line through the training data exactly, it is to **do well on unseen test cases!**



Recap from last class:

What can go wrong in linear regression

Two main categories of fixes:

1. Remove features until the problem is well-behaved (“feature selection”).
2. Leave the features as they are, but *add constraints to the system* to “tighten it up” (aka “regularization”)—next topic!

$$\hat{y} = E_Y[p(y|x)] = \textcolor{brown}{w}^T x + \textcolor{blue}{w}_0, \text{ for } w, x \in \mathbb{R}^d$$

L2 regularized linear regression

To *shrink* $\mathbf{w}$ to have smaller norm than the MLE solution, we add a “penalty” term to the loss function:

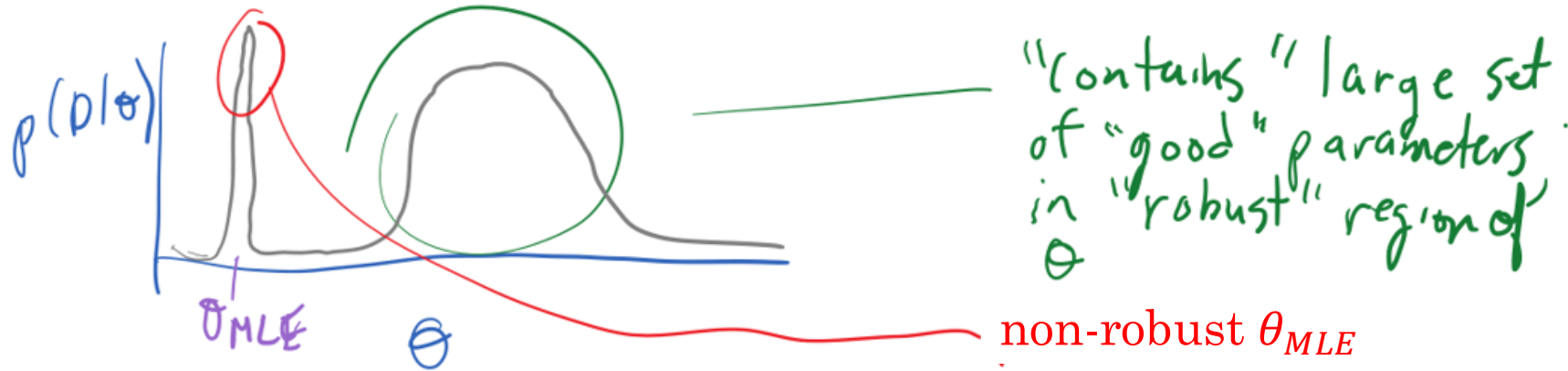
$$\mathcal{L} = (\mathbf{y} - A\mathbf{w})^T (\mathbf{y} - A\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$
$$\mathbf{w}_{L_2} = \operatorname{argmin}_{\mathbf{w}} (\mathbf{y} - A\mathbf{w})^T (\mathbf{y} - A\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

Also called “Ridge” regression, or L2 linear regression.
Related to *Bayesian* modeling (next).

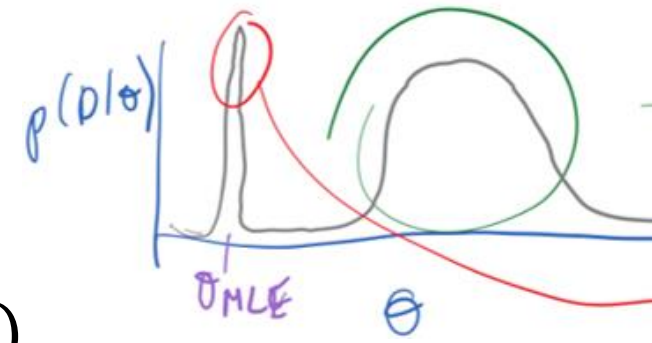
Recall how MLE can go wrong?

MLE yields a "point estimate" of our parameter

- When we perform MLE, we get just one single estimate of the parameter, θ , rather than a distribution over it which captures uncertainty.
- In Bayesian statistics, we obtain a (posterior) distribution over θ . We will touch more on this in a few lectures.



The Bayesian modelling approach



- Bayesians put a *prior distribution* on the parameters, $p(\theta)$.
- Then they seek to compute the *posterior distribution*, $p(\theta|D)$.
- Then, predictive distribution is given by

$$p(y|x) = \int_{\theta} p(y|x, \theta) p(\theta|D) d\theta = E_{\theta}[p(y|x, \theta)]$$

- Procedurally, this is done using Bayes' rule:

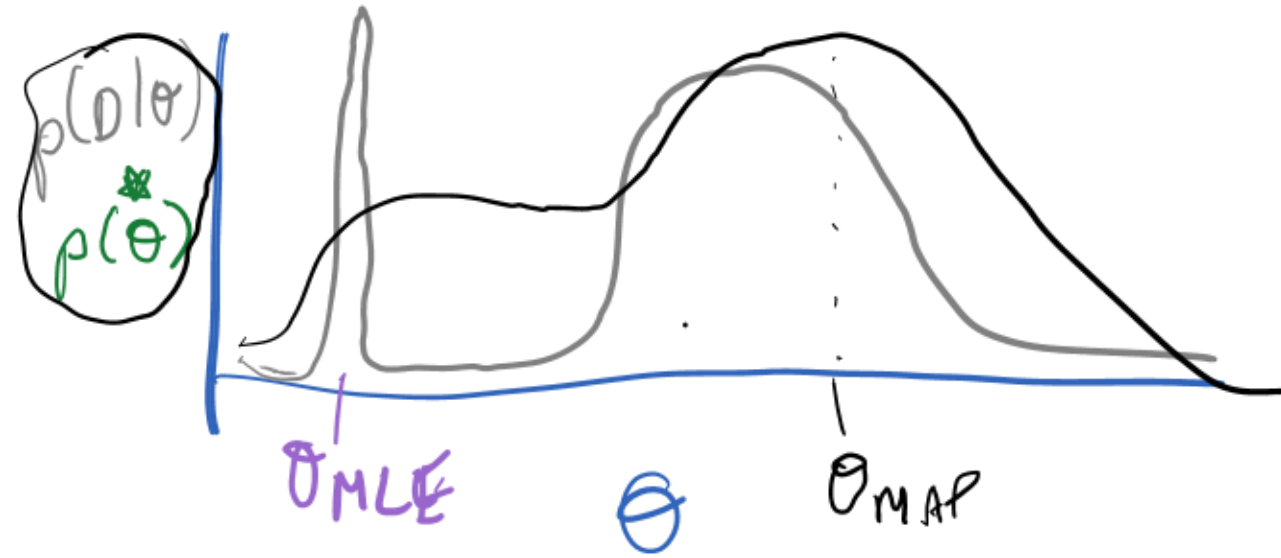
$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.$$

- Difficult in practice! $p(D) = \int_{\theta} p(D, \theta) d\theta = \int_{\theta} p(D|\theta)p(\theta) d\theta$
- We will be lazy, instead being fake Bayesians, yielding L2 regression:
- $\theta_{lazy} = \operatorname{argmax}_{\theta} p(\theta|D)$ *Maximum A Posteriori (MAP) estimation.*

MAP: the lazy Bayesian (*Maximum A Posteriori*)

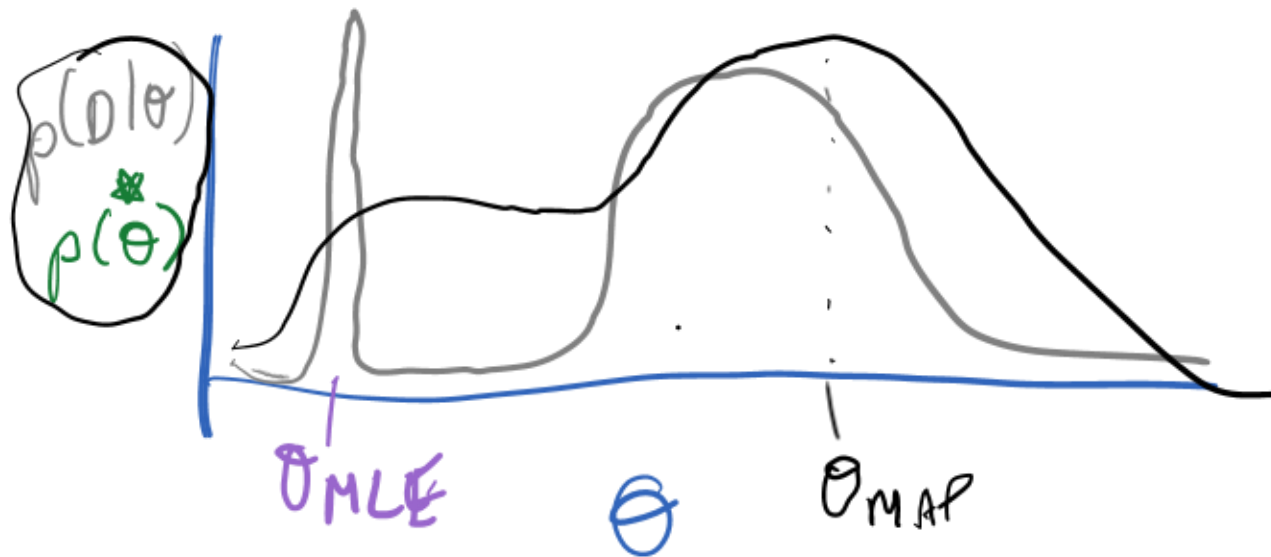
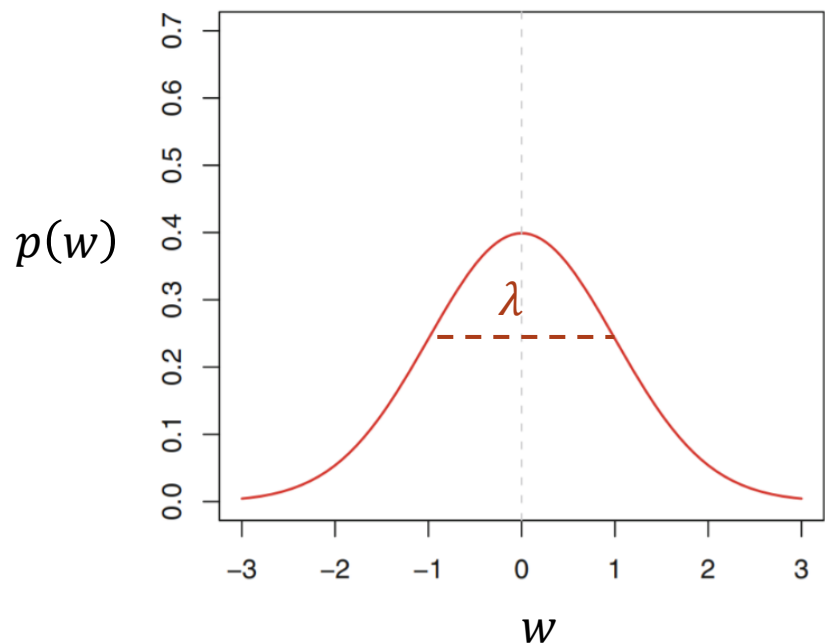
- Still use a prior over parameters, $p(\theta)$.
- Finds point estimate of the parameter that maximizes the posterior.
- $\theta_{MAP} = \operatorname{argmax}_{\theta} p(\theta|D)$

$$\begin{aligned} &= \operatorname{argmax}_{\theta} \frac{p(D|\theta) \cancel{p(D)}}{\cancel{p(D)}} \\ &= \operatorname{argmax}_{\theta} p(D|\theta) p(\theta) \end{aligned}$$



A prior for small weights yields L2 regression!

- Zero-mean prior, $p(w) = N(w; 0, \lambda I)$.
- Bayesian posterior, $p(w|D) = \frac{p(D|w)p(w)}{p(D)}$ is then “nice” in that everything is Gaussian (can analytically work it out using MVGs).



MAP for linear regression w Gaussian prior

$$\begin{aligned}w_{MAP} &= \operatorname{argmax}_w \log p(D|w) p(w) = \operatorname{argmax}_w \log p(D|w) + \log N(w; 0, \lambda I) \\&= \operatorname{argmax}_w \sum_{i=1}^N \log N(y_i | w^T x_i, \sigma^2) + \log N(w | 0, \lambda I) \\&= \operatorname{argmin}_w + \frac{1}{2\sigma^2} (y - Aw)^T (y - Aw) - \sum_{i=1}^d \log \left[\frac{1}{\sqrt{2\pi\lambda}} \exp \left(-\frac{(w_i - 0)^2}{2\lambda} \right) \right] \\&= \operatorname{argmin}_w \frac{1}{2\sigma^2} (y - Aw)^T (y - Aw) + \sum_{i=1}^d \left[-\log \frac{1}{\sqrt{2\pi\lambda}} + \frac{w_i^2}{2\lambda} \right] \\&= \operatorname{argmin}_w \frac{1}{2\sigma^2} (y - Aw)^T (y - Aw) + \sum_d \frac{1}{2\lambda} w_i^2 \\&= \operatorname{argmin}_w \frac{1}{2\sigma^2} (y - Aw)^T (y - Aw) + \frac{1}{2\lambda} \|w\|_2^2 \\&= \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \cancel{2\sigma^2} \frac{1}{2\lambda} \|w\|_2^2 \\&= \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda' \|w\|_2^2 \quad \text{for } \lambda' = \frac{\sigma^2}{\lambda}.\end{aligned}$$

Equivalence between
MAP w Gaussian prior
and L2 regression!

Obtaining the MAP/L2 solution

$$\begin{aligned} w_{L_2} &= \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda \|w\|_2^2 \\ &= \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda w^T w \end{aligned}$$

Take partial derivative and set to zero:

$$\nabla_w \mathcal{L}_{MAP} = -2A^T y + 2A^T A w + 2\lambda I w$$

$$\rightarrow 0 = -A^T y + A^T A w + \lambda I w$$

$$\rightarrow A^T y = (A^T A + \lambda I) w$$

$$\rightarrow (A^T A + \lambda I)^{-1} A^T y = w$$

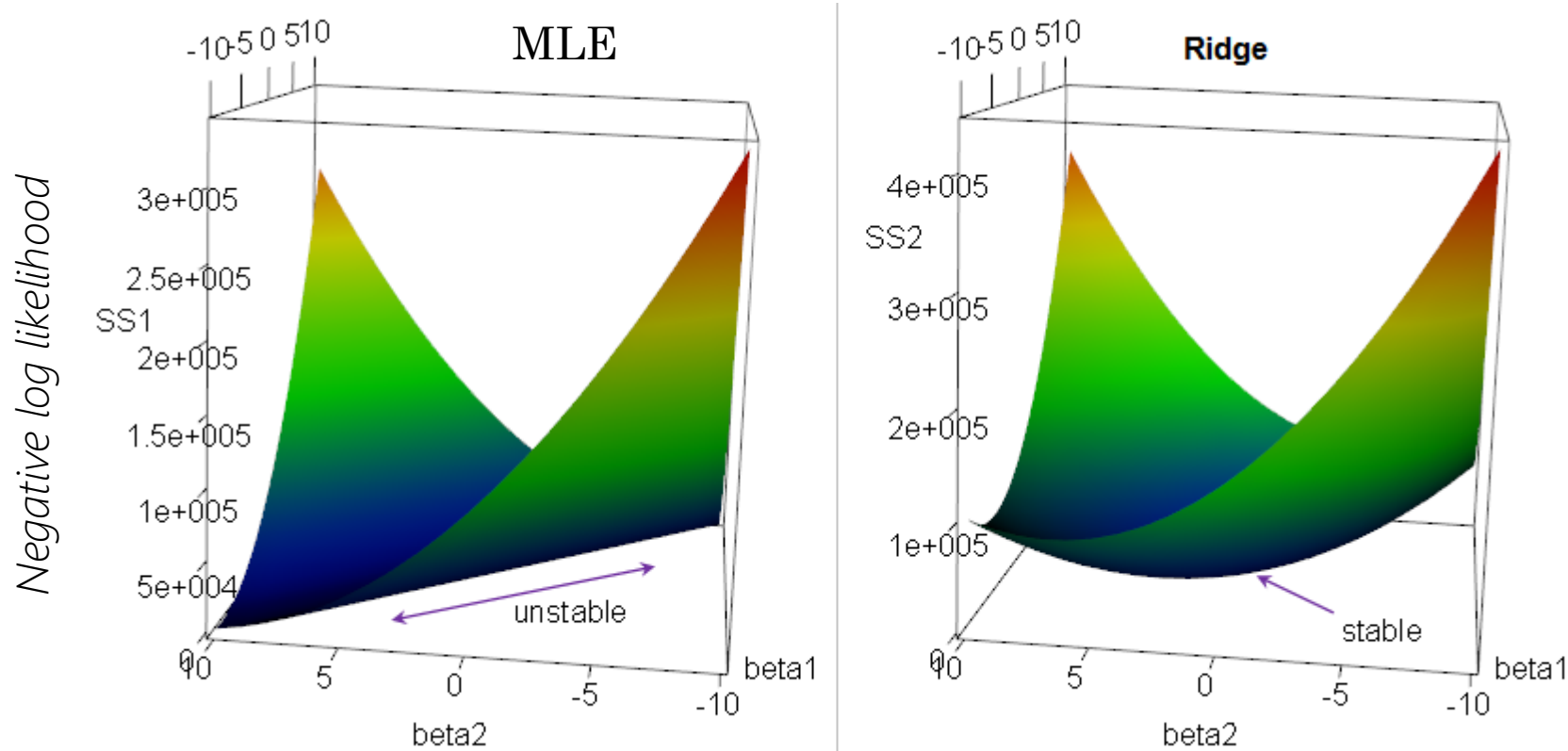
$$\text{So } w_{L_2} = (A^T A + \lambda I)^{-1} A^T y.$$

If $\lambda > 0$, we can invert $(A^T A + \lambda I)$.

$$\begin{aligned} Z &= \Phi D \Phi^T \\ Z^{-1} &= \Phi D^{-1} \Phi^T \\ \text{where } D^{-1} &= \begin{bmatrix} \lambda_1^{-1} & & \\ & \lambda_2^{-1} & \\ & & \ddots \\ & & & \lambda_n^{-1} \end{bmatrix} \end{aligned}$$

Bending the “flat ridge” to get one optimal w

When some features are linearly dependent (can't invert $A^T A$), we have ∞ many equally good solutions that form a ridge.

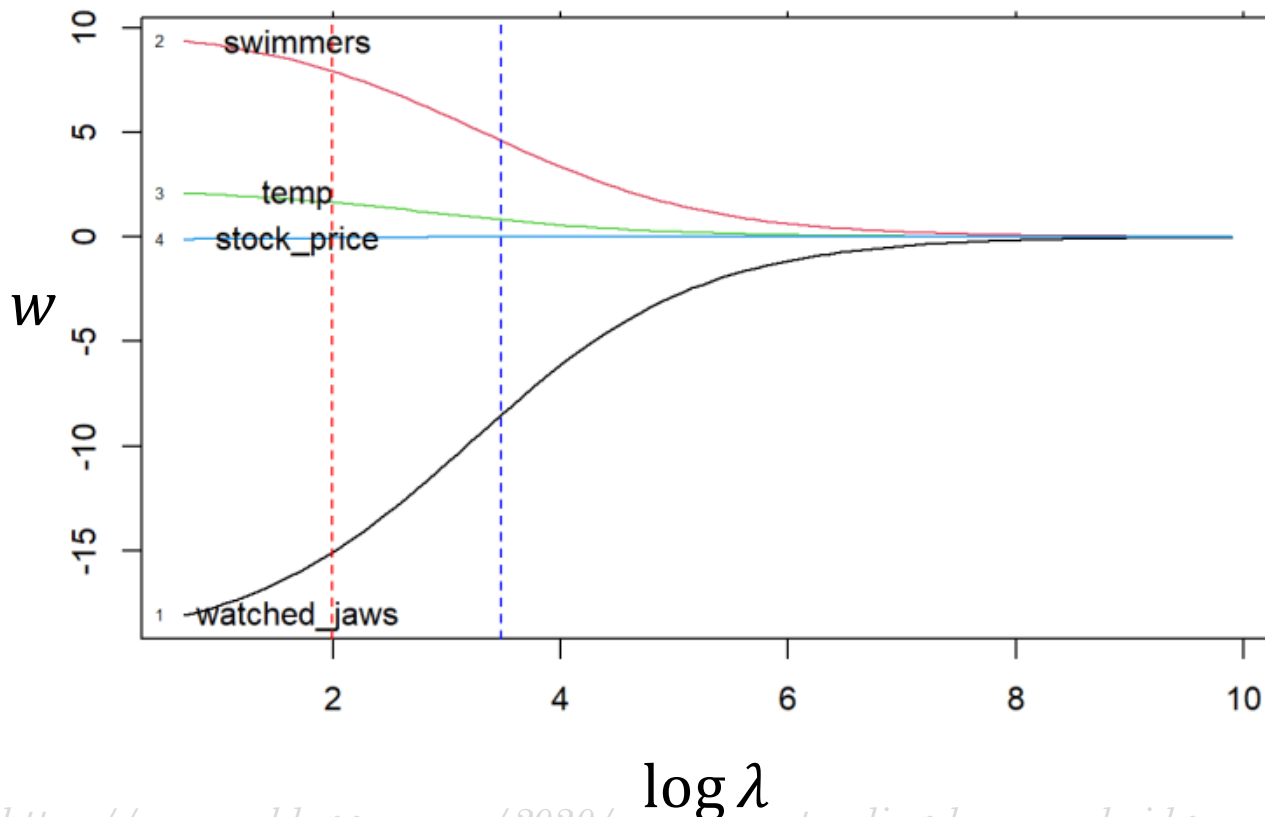


$$w_{L_2} = (A^T A + 0)^{-1} A^T y.$$

$$w_{L_2} = (A^T A + \lambda I)^{-1} A^T y.$$

Effect of value of λ

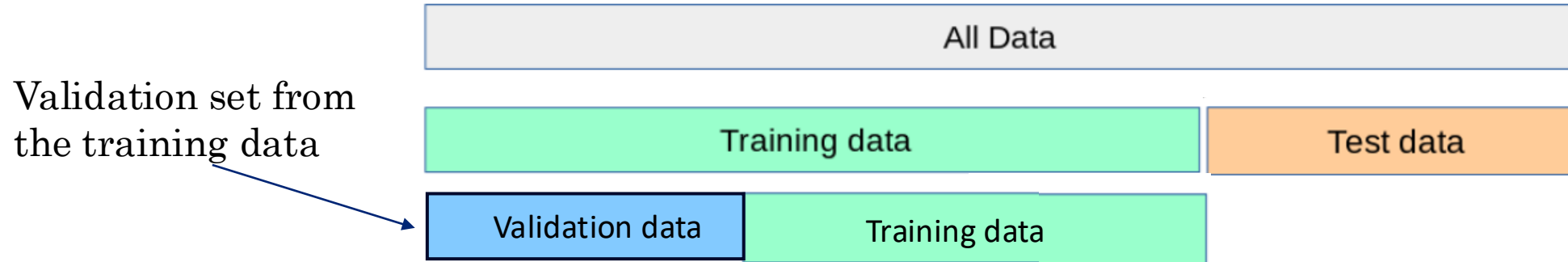
$$\mathcal{L}_{MAP} = (y - Aw)^T (y - Aw) + \lambda \|w\|_2^2$$



- Practically, how should we set λ ?
- Can we treat it as a parameter in the loss, and minimize wrt it?
- No: cannot use MLE!
- Need independent data, a *validation set* on which to evaluate the loss.

Train/validation/test split

$$\mathcal{L}_{MAP} = (y - Aw)^T (y - Aw) + \lambda \|w\|_2^2$$

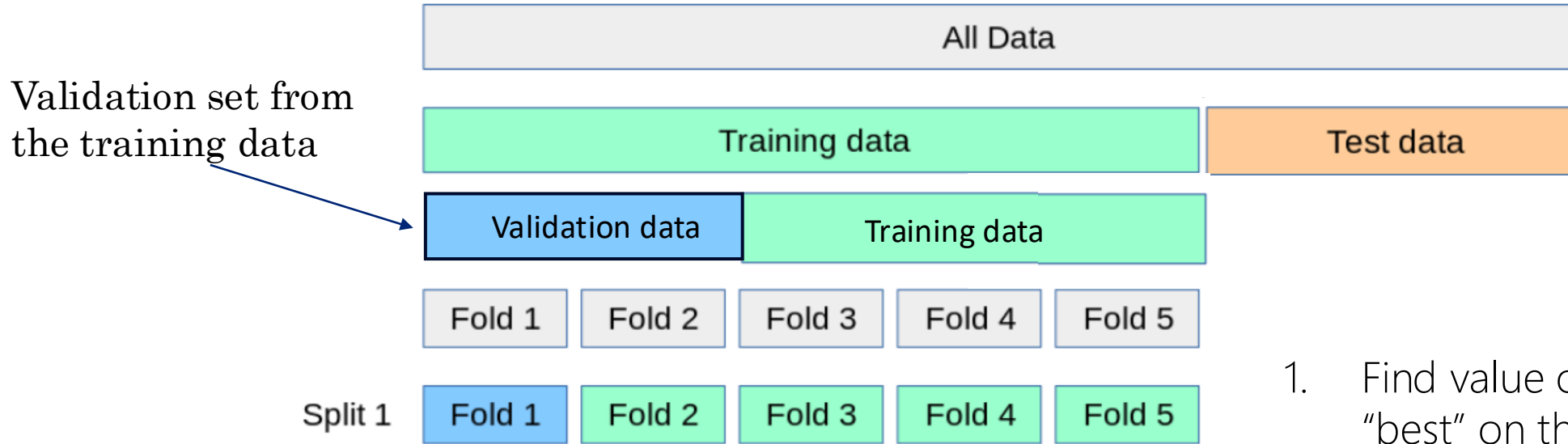


1. Find value of hyperparameter that is “best” on the **validation set**.
2. Quantify performance on the **test data**.

“Best”/quantify require computing e.g., log likelihood on the held out data, and this is our metric.

Train/validation/test split

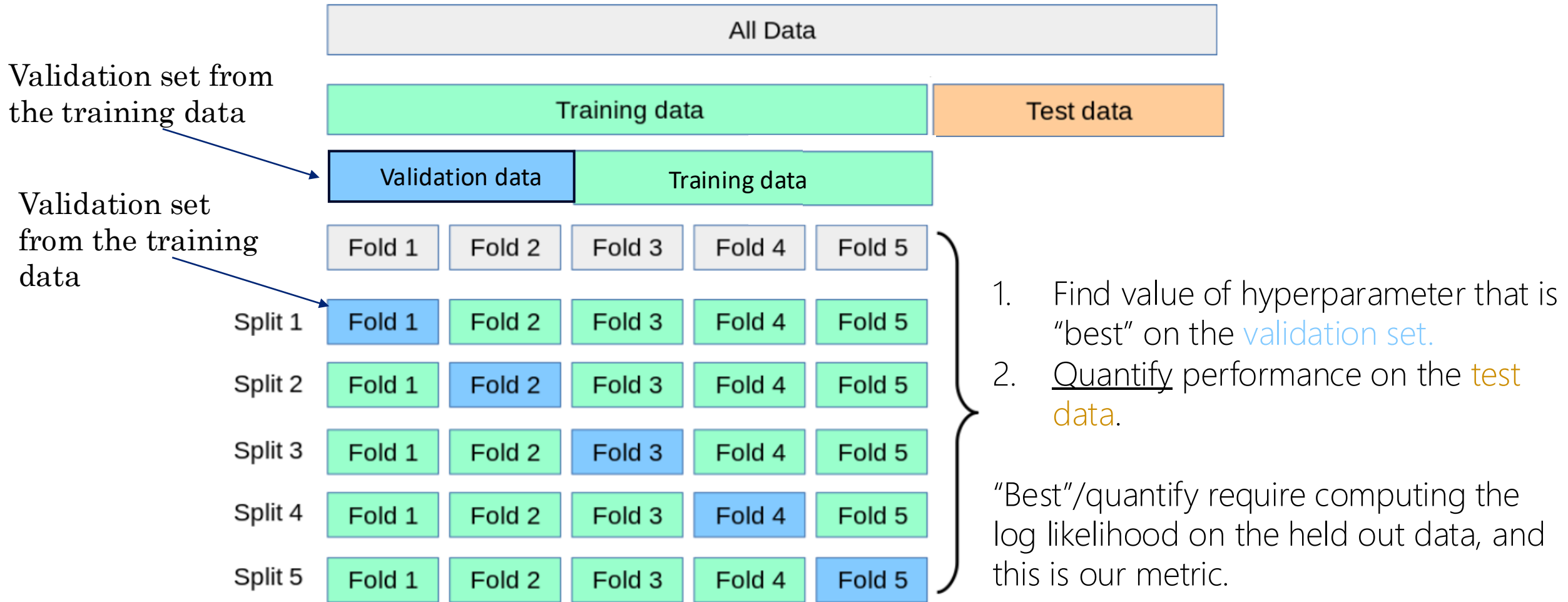
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1. Find value of hyperparameter that is "best" on the **validation set**.
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K-fold cross-validation

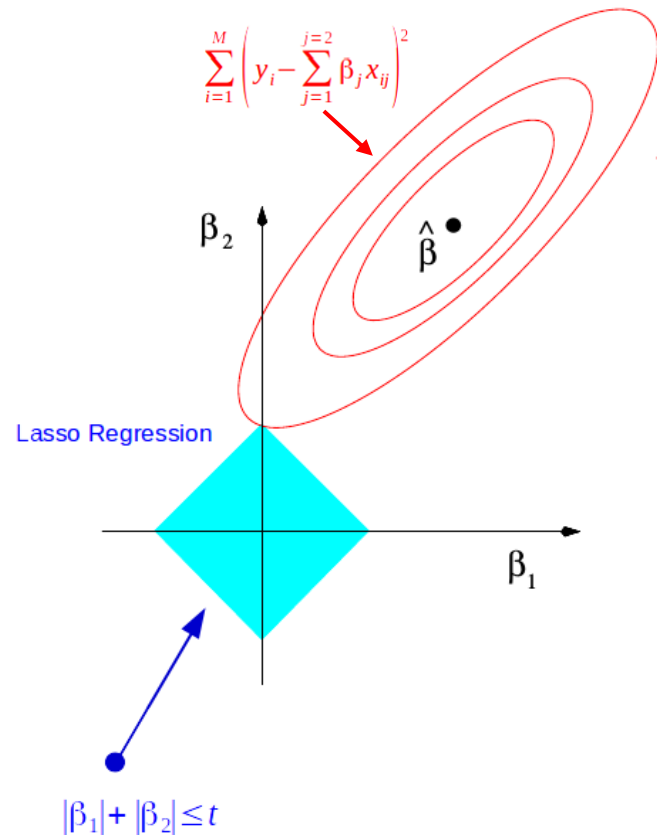


Other priors for MAP in linear regression?

- What if wanted to have the linear function only depend on a few features (*i.e.*, the coefficients of $\mathbf{w}$ are sparse)?
- Add a penalty that counts the # of non-zero weights—an L_0 penalty, $\lambda \|\mathbf{w}\|_0$.
- Not differentiable! (becomes combinatorial optimization—nasty).
- But...the L_1 norm penalty, $\lambda \|\mathbf{w}\|_1 = \lambda \sum_d |w_d|$, tends to induce sparse $\mathbf{w}$ —differentiable everywhere except at $\mathbf{w}_i = \mathbf{0}$ which can be worked around.
- This is called *Lasso* regression, or L_1 -penalized regression.

L_1 -penalized linear regression, aka *Lasso*

- $w_{L_1} = \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda \|w\|_1$
- Why does the L_1 norm penalty tends to induce sparse w ?
- Equivalent to MLE with constraint $\|w\|_1 < \text{constant}$.
- "Pointy" constraint surface is jutting out along the axes.
- In many cases, the L_1 norm constraint will cause the unconstrained solution to intersect the constraint at a corner.
- The corners are where some coefficients are 0, which is a sparse solution..



The p -norm of a vector $\mathbf{x} \in \mathbb{R}^d$ is defined as:

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$$

for $p \geq 1$.

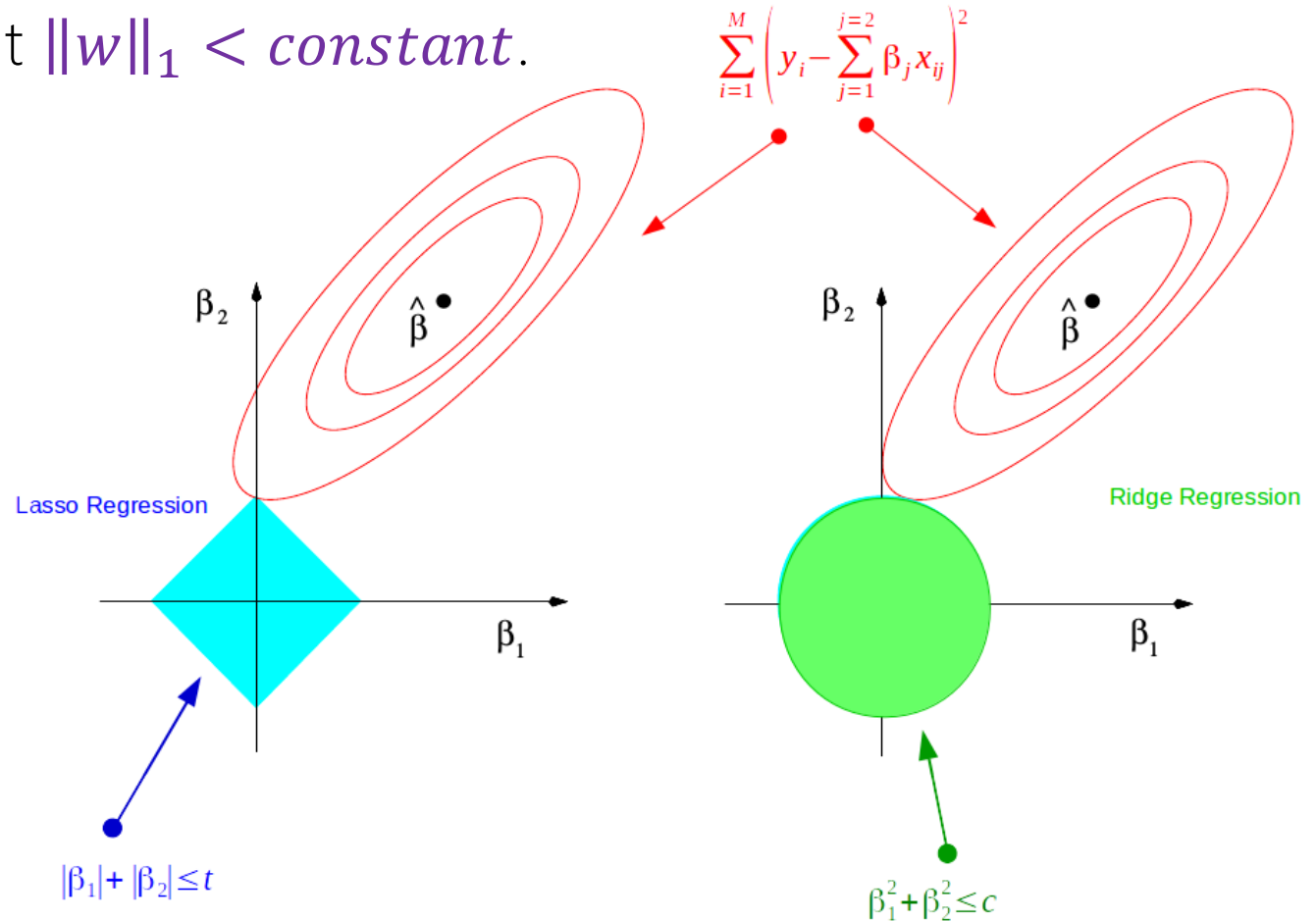
Common cases:

- $p = 1$ (Manhattan/taxicab norm): $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \dots + |x_d|$
- $p = 2$ (Euclidean norm): $\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$
- $p = \infty$ (max norm): $\|\mathbf{x}\|_\infty = \max_i |x_i|$

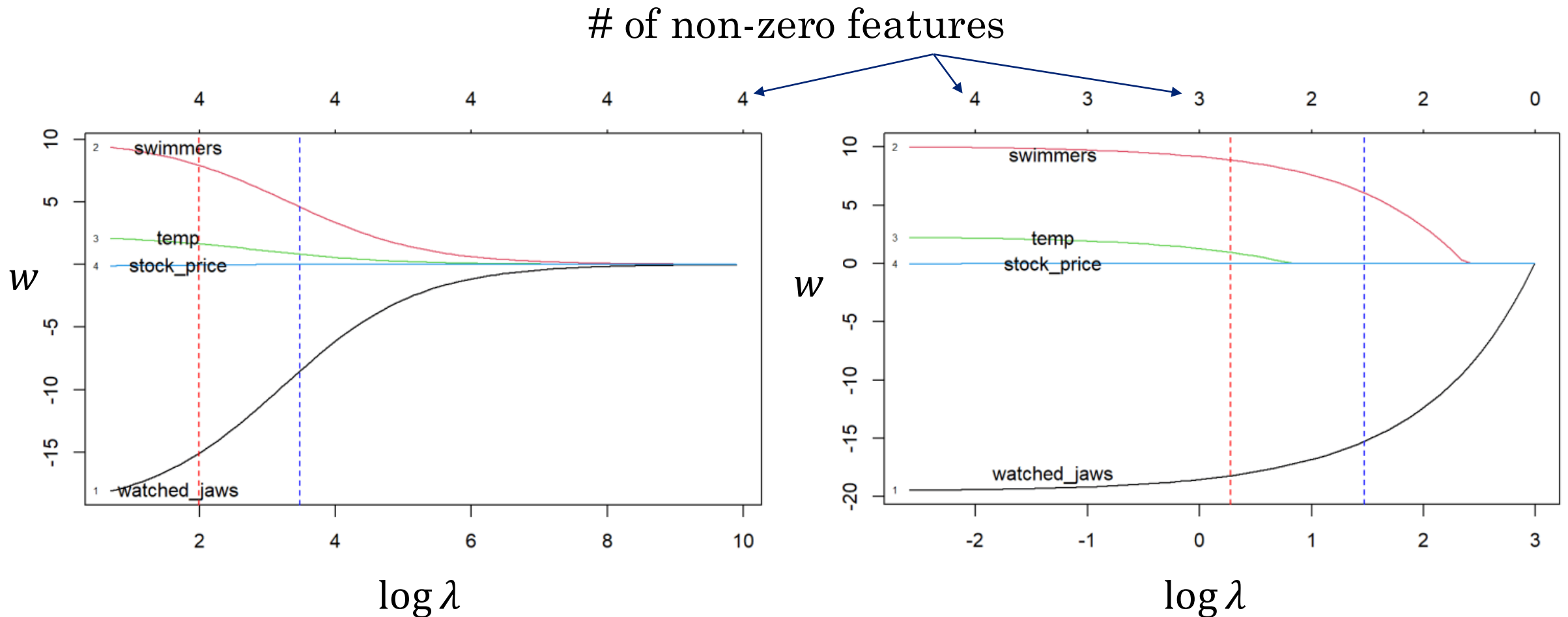
The 2-norm is what we typically mean when we just write $\|\mathbf{x}\|$ without a subscript.

L_1 -penalized linear regression, aka *Lasso*

- $w_{L_1} = \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda \|w\|_1$ $w_{L_2} = \operatorname{argmin}_w (y - Aw)^T (y - Aw) + \lambda \|w\|_2^2$
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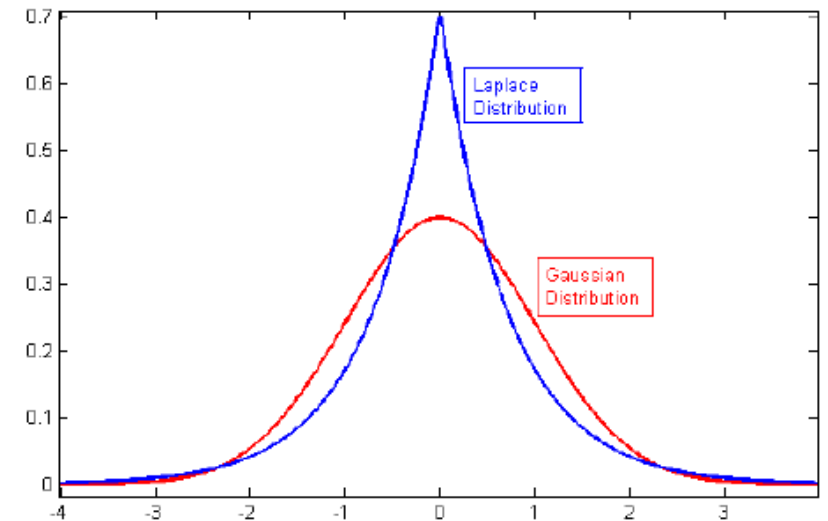
Ridge vs Lasso: shrinkage vs sparsity



MAP interpretation for Lasso/ L_1 -penalized linear regression?

- L_2 regression arose from a $N(0, \lambda I)$ prior.
- Is there a prior corresponding to L_1 ?
- Yes, the Laplace prior!

$$p(w) = \frac{1}{2\lambda'} \exp\left(-\frac{\|w\|_1}{\lambda'}\right).$$

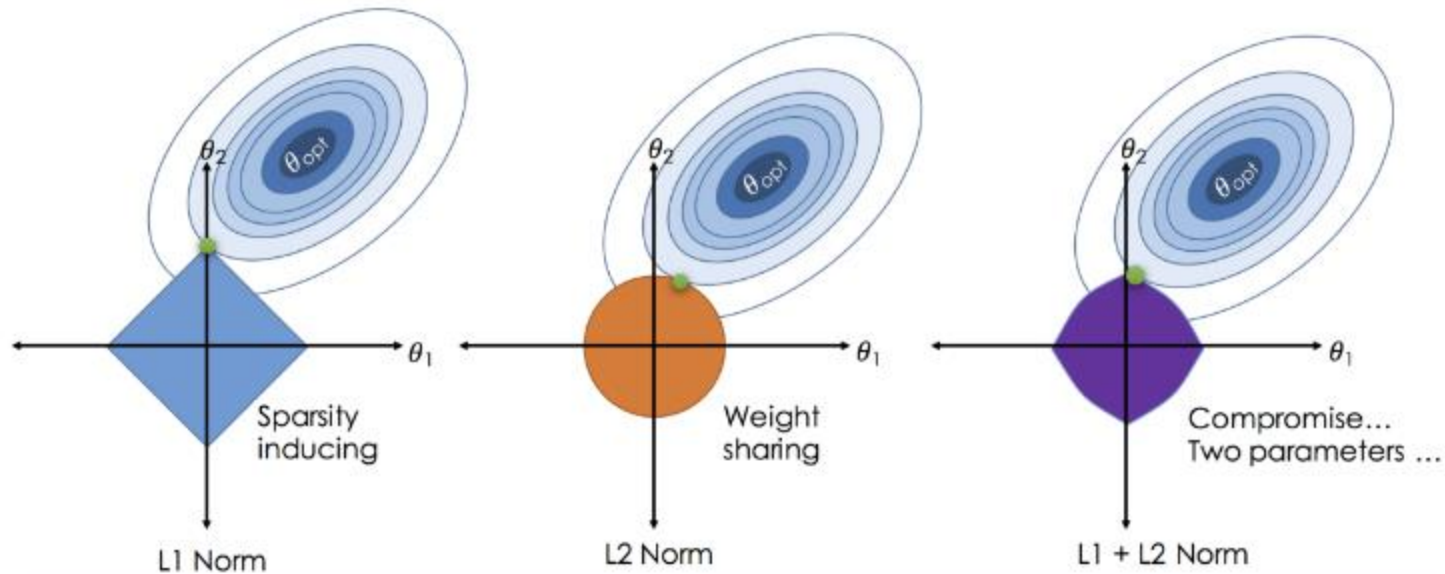


Issues with LASSO:

- If highly correlated features, tends to ignore all but one.

Combine L_1 and L_2 penalties?

Yes, "elastic net regression".



A thought experiment

Consider:

- Use MLE on data, $D = \{x_i, y_i\}$ to get $\hat{y}_i = p_{\theta}(y|\Phi(x))$ using linear regression.
- Assume an abundance of data (millions of data points), and only 100 parameters.
- Suppose get accuracy $\mp \$1000$ of sale price when applying to held out part of our data.
- Can we assume this model will get $\mp \$1000$ on any test set that may come in the future?

Actual vs. predicted sale price of house

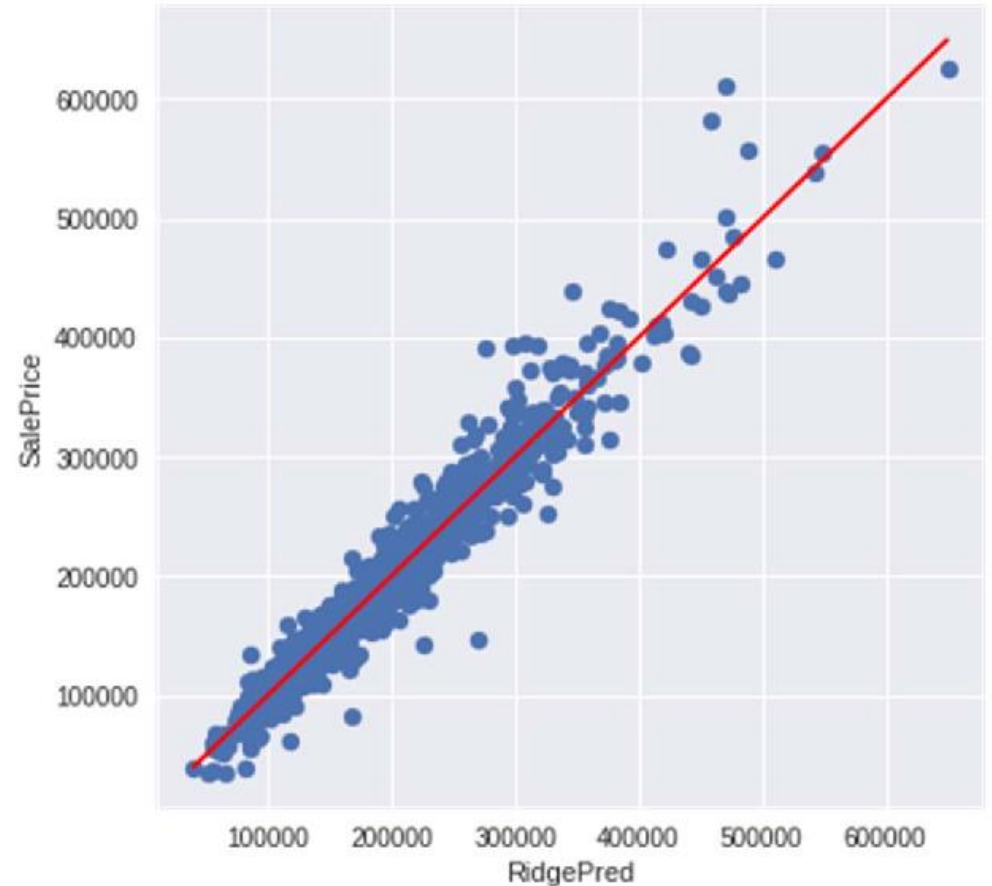


Fig. 4 Ridge Prediction for Training Data.

Causation vs correlation

Breakingviews

Zillow's failed house flipping

Reuters

WSJ NOV. 2021 : *“The company expects to record losses of more than \$500 million from home-flipping by the end of this year and is laying off a quarter of its staff.”*

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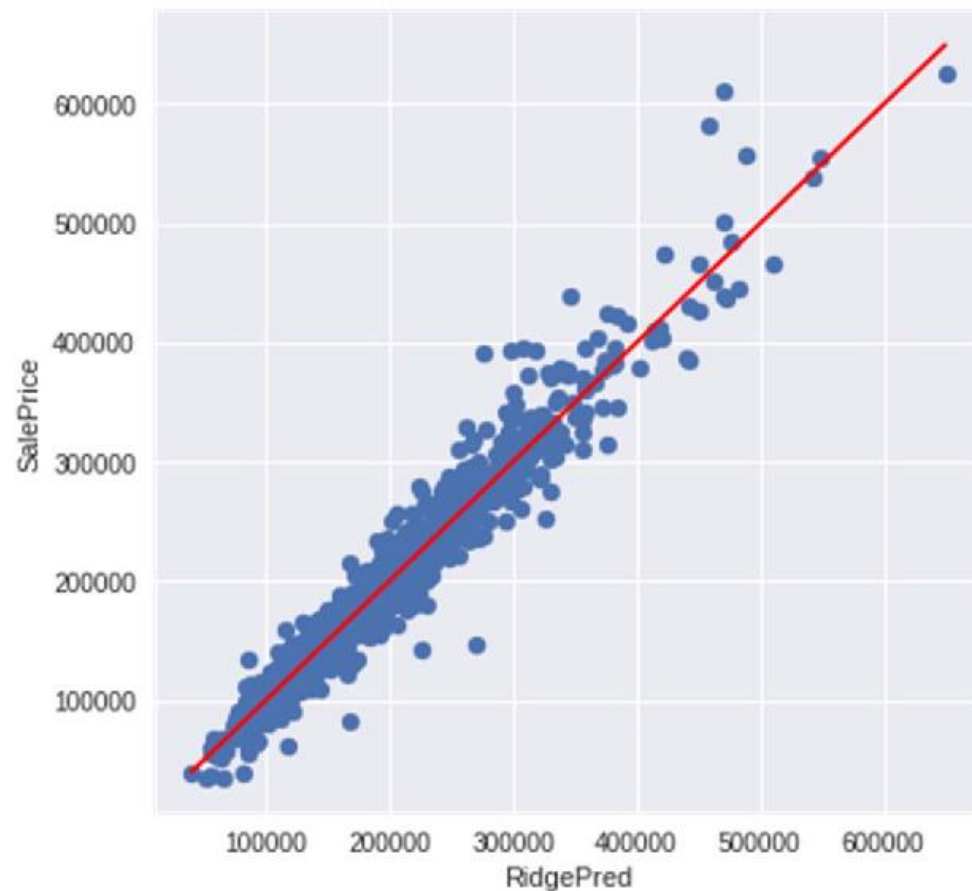


Fig. 4 Ridge Prediction for Training Data.